HACKATHON — SYNTHETIC DATA SETS

Three industry-specific databases with realistic, correlated synthetic data. Each database has already been deployed to your lab account and is ready to query.

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1. Financial Services & Insurance (`FINSERV_DW`)

File: `01_financial_services_insurance.sql` | **~480K records** | **15 companies**

A full retail banking and insurance data warehouse covering customer accounts, transactions, insurance policies, claims, risk assessments, and fraud detection.

Schemas & Key Tables:

Schema	Tables	Description
`CORE`	`COMPANIES` , `CUSTOMERS` , `ACCOUNTS`	15 financial firms, 50K customers (VIP/Premium/Standard/Basic tiers), ~85K accounts
`TRANSACTIONS`	`TRANSACTIONS`	200K transactions across 6 types (debit, credit, transfer, fee, interest, dividend) with realistic merchant names and amount distributions
`INSURANCE`	`POLICIES` , `CLAIMS`	35K policies (auto, home, life, health, umbrella, commercial) and ~14K correlated claims
`RISK`	`RISK_ASSESSMENTS` , `FRAUD_ALERTS`	50K risk scores inversely correlated with credit scores, 8K fraud alerts weighted toward high-risk customers
`ANALYTICS`	`CUSTOMER_360` , `POLICY_PERFORMANCE`	Pre-built views for customer lifetime value and claims ratio analysis

Built-in Correlations: - VIP customers have 3–6 accounts, Basic customers have 1 - VIP accounts receive ~35% of all transactions despite being ~13% of accounts - CRITICAL-risk policies have a ~60% claim rate vs ~8% for LOW-risk - Fraud alerts concentrate on customers with risk scores above 40 - Claims `reported_date` always follows `incident_date`

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2. Healthcare & Life Sciences (`HEALTHCARE_DW`)

File: `02_healthcare_life_sciences.sql` | **~480K records** | **12 organizations**

A clinical and administrative healthcare data warehouse spanning patient records, encounters, diagnoses, prescriptions, billing, and clinical research.

Schemas & Key Tables:

Schema	Tables	Description
`CORE`	`ORGANIZATIONS`, `PROVIDERS`, `PATIENTS`	12 healthcare orgs, 5K providers with specialties, 60K patients with demographics and insurance
`CLINICAL`	`ENCOUNTERS`, `DIAGNOSES`	~180K encounters (outpatient, office visit, ER, inpatient, telehealth) and ~150K diagnoses using real ICD-10 codes
`PHARMACY`	`MEDICATIONS`, `PRESCRIPTIONS`	20 common medications mapped to diagnoses, ~100K prescriptions with dosage and cost
`BILLING`	`CLAIMS`, `PAYMENTS`	~120K billing claims with cost correlated to encounter type, plus payment breakdowns
`RESEARCH`	`CLINICAL_TRIALS`, `TRIAL_ENROLLMENTS`	500 clinical trials across 10 therapeutic areas with 12K patient enrollments
`ANALYTICS`	`PATIENT_360`, `DIAGNOSIS_TREATMENT_MAP`	Pre-built views for patient utilization and diagnosis-to-prescription mapping

Built-in Correlations: - Chronic patients (diabetes, hypertension, etc.) have 5–12 encounters vs 1–3 for healthy patients - Older patients receive age-appropriate diagnoses (heart disease, osteoarthritis for 65+; asthma, anxiety for younger) - Diagnoses map to correct medications (diabetes → metformin, hypertension → lisinopril) - ER and inpatient encounters cost significantly more than outpatient/telehealth - Prescription costs and billing amounts follow realistic healthcare ranges

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3. Automotive & Manufacturing (`AUTO_MFG_DW`)

File: `03_automotive_manufacturing.sql` | **~400K records** | **12 companies**

An end-to-end automotive manufacturing data warehouse covering vehicle models, supply chain, dealer sales, quality control, warranty claims, and production workforce.

Schemas & Key Tables:

Schema	Tables	Description
`PRODUCT`	`COMPANIES`, `VEHICLE_MODELS`, `PARTS_CATALOG`	12 companies (4 OEMs, 8 suppliers), 150 vehicle models with specs (ICE/HEV/PHEV/BEV), 3K parts
`SUPPLY_CHAIN`	`SUPPLIERS`, `PURCHASE_ORDERS`	400 suppliers with quality ratings and tiers, 50K purchase orders
`SALES`	`DEALERS`, `CUSTOMERS`, `VEHICLE_SALES`	1.5K dealers, 80K customers, 100K vehicle sales with financing details
`QUALITY`	`INSPECTIONS`, `DEFECTS`, `WARRANTY_CLAIMS`	25K inspections, 8K defects, 30K warranty claims tied to vehicle age

Schema	Tables	Description
`WORKFORCE`	`EMPLOYEES`, `PRODUCTION_SHIFTS`	12K employees across departments, 50K shift records with production metrics
`ANALYTICS`	`SALES_BY_MODEL`, `SUPPLIER_SCORECARD`, `WARRANTY_BY_MODEL`	Pre-built views for sales performance, supplier quality, and warranty cost analysis

Built-in Correlations: - Fleet customers average ~4 purchases vs ~0.6 for new customers - Lower-quality-rated suppliers generate disproportionately more defects (bottom 25% of suppliers → 50% of defects) - Day shifts produce more units (180–280) than night shifts (100–200) - Warranty claims are distributed across bumper-to-bumper, powertrain, and extended coverage - Purchase order delivery dates always follow order dates (14–90 day lead times)